

ORIGINAL CONTRIBUTION

Association of Successful Recanalization and Functional Outcomes in Minor Ischemic Stroke With Proven Occlusion: A Secondary Analysis of TEMPO-2 Trial

Nishita Singh¹, MD; Daniel Strbian², MD; Shabnam Vatanpour³, MSc; Thalia S. Field⁴, MD; Amy Y.X. Yu⁵, MD; Aravind Ganesh⁶, MD; Philip A. Barber⁷, MD; Philip M.C. Choi⁸, MD; Brian H. Buck⁹, MD; Timothy J. Kleinig¹⁰, MD; Bruce Campbell¹¹, MD; Carlos Molina¹², MD; Keith W. Muir¹³, MD; Michael D. Hill¹⁴, MD; Shelagh B. Coutts¹⁵, MD

BACKGROUND: Complete recanalization of cerebral arteries is strongly associated with good functional outcome in ischemic stroke. We hypothesize that successful recanalization results in better functional outcomes.

METHODS: This is a secondary observational cohort analysis of TEMPO-2 (Tenecteplase Versus Standard of Care for Minor Ischemic Stroke With Proven Occlusion), a randomized controlled trial comparing tenecteplase with standard of care (control) in minor stroke (National Institutes of Health Stroke Scale score ≤ 5) with intracranial occlusion/focal perfusion abnormality ≤ 12 hours of onset. Among those enrolled based on computed tomography angiography with visible occlusion, a follow-up computed tomography angiography was done at 4 to 8 hours after randomization. The primary outcome was return to baseline functional outcomes using the modified Rankin Scale score at 90 days. Safety outcomes included stroke progression (National Institutes of Health Stroke Scale score ≥ 2 worsening), bleeding events, and mortality. Patients with successful recanalization, defined as revised Arterial Occlusive Lesion score ≥ 2 b/3, were compared with those with unsuccessful recanalization on follow-up computed tomography angiography. Regression analysis was used to assess the association of successful recanalization with outcomes after adjusting for age, sex, baseline stroke severity, and onset-to-randomization time.

RESULTS: Of the 886 enrolled patients, 517 (58.3%) with follow-up computed tomography angiography were included. Of these, 178 (34.6%) had successful recanalization (122 [68.5%]: tenecteplase, 56 [31.5%]: control), and 336 (65.4%) did not achieve successful recanalization (unsuccessful recanalization; 134: tenecteplase, 202: control). Baseline characteristics were similar between patients with and without successful recanalization. Successful recanalization was significantly associated with the primary outcome as compared with unsuccessful recanalization (adjusted risk ratio, 1.21 [95% CI, 1.07–1.34]). Patients with successful recanalization had significantly lower rates of stroke progression as compared with unsuccessful recanalization (2.8% versus 13.1%, adjusted risk ratio, 0.21 [95% CI, 0.08–0.52]). Multivariable analysis showed that tenecteplase treatment was the strongest independent predictor of successful recanalization (odds ratio, 3.48 [95% CI, 2.33–5.18]).

CONCLUSIONS: Successful recanalization is a critical determinant of early and 90-day functional recovery in patients with minor stroke with intracranial occlusion, regardless of treatment modality. Tenecteplase significantly increases the odds of achieving successful recanalization compared with standard care.

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GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: angiography ■ cerebral arteries ■ ischemic stroke ■ tenecteplase ■ tomography

Correspondence to: Nishita Singh, MD, Section of Neurology, Health Sciences Centre, GF543, Sherbrook St, Winnipeg, MB R3A 1R9. Email nishitaneurology@gmail.com
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Nonstandard Abbreviations and Acronyms

CT	computed tomography
CTA	computed tomography angiography
EQ-5D-5L	European Quality of Life score index
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Severity Scale
OR	odds ratio
QOL	quality of life
VAS	visual analog scale

Up to 50% of patients with ischemic stroke initially present with symptoms that are nondisabling.¹ Evidence of an intracranial occlusion defines a subpopulation at high risk for early neurological deterioration, most often occurring within the first 24 hours after presentation.² This is true even if the deficits initially resolve.^{2,3} Fast and complete recanalization of cerebral arteries is a strong predictor of good functional outcomes in stroke broadly, and this may similarly apply to minor strokes with proven intracranial occlusion.⁴ In this subgroup, symptoms may initially be mild or nondisabling, precluding thrombolytic treatment, but persistent vessel occlusion may lead to worsening deficits, infarct growth, and later clinical deterioration when thrombolytic treatment is contraindicated. This concept was the fundamental premise behind the TEMPO-2 trial (Tenecteplase Versus Standard of Care for Minor Ischemic Stroke With Proven Occlusion) that thrombolytic treatment would lead to greater recanalization, which in turn would lead to more favorable outcomes at 90 days. The results of the TEMPO-2 trial showed no benefit and possible harm from treatment with intravenous tenecteplase in patients with minor stroke with intracranial occlusion or perfusion mismatch.⁵

Although thrombolytic agents like tenecteplase are known to enhance early recanalization, contemporary standard medical management, including early initiation of dual antiplatelet therapy, may also facilitate endogenous thrombolysis. However, there are currently no clinical trial data directly comparing rates of recanalization achieved with contemporary medical therapy versus thrombolysis in this population.

Using data from the TEMPO-2 randomized controlled trial, which compared tenecteplase with nonthrombolytic control in patients with minor stroke with intracranial occlusion, we assessed predictors of successful recanalization and early neurological improvement.

METHODS

Data collected for the study, including deidentified individual participant data and a data dictionary defining each field in the set, can be made available to others on reasonable request

and after signing appropriate data sharing agreements. Such requests must be approved by all the respective ethics boards and appropriate data custodians.

Study Design

The TEMPO-2 trial was an investigator-initiated, multicenter, prospective, randomized, open-label with blinded end point assessment (PROBE [Prospective Randomized Open Blinded End-Point]), parallel group, controlled trial, designed to test the superiority of intravenous tenecteplase (0.25 mg/kg) over nonthrombolytic standard of care in patients with minor ischemic stroke deficits, defined as National Institutes of Health Stroke Scale (NIHSS) score 0 to 5, and intracranial occlusion or focal perfusion lesion within 12 hours from onset of symptoms. Patients with prestroke modified Rankin Scale score >2 were not eligible for the trial. The trial was conducted at 48 hospitals in Australia, Austria, Brazil, Canada, Finland, Ireland, New Zealand, Singapore, Spain, and the United Kingdom. The methods of this trial have been previously published.⁶ Informed consent was obtained from all participants, and the study was approved by all local institutions. We examined only those patients who were enrolled based on computed tomography angiogram (CTA) with visible occlusion. This is a secondary observational cohort analysis comparing patients with and without successful recanalization in patients enrolled in the TEMPO-2 trial who had a baseline CTA.

Imaging Analysis

All patients received noncontrast head computed tomography (CT), CTA, or CT perfusion as baseline imaging per standard of care before randomization. Recanalization was assessed on a 4- to 8-hour CTA in patients with baseline occlusion, per study protocol. Patients enrolled on the basis of a CT perfusion-defined perfusion deficit did not have follow-up CTA. Successful recanalization was defined as revised Arterial Occlusive Lesion score $\geq 2b/3$ on follow-up CTA. An independent imaging core laboratory, including authors with >5 years of experience, read all imaging data (baseline noncontrast CT, CTA, CT perfusion, and follow-up scans). Core laboratory members were blinded to treatment allocation and outcome. The collection and assessment of the imaging data were reviewed in consensus readings. Disagreements between 2 readers were resolved by an experienced independent third reader.

Outcomes

We explored prespecified outcomes by successful recanalization status, defined as revised Arterial Occlusive Lesion score $\geq 2b/3$; yes/no) for the following outcomes:

- Functional disability: The primary functional outcome was defined as return to baseline neurological functioning as measured by the modified Rankin Scale (mRS) score, using a sliding dichotomy approach. A responder was defined as follows: (1) if the premorbid mRS score was 0 to 1, then the mRS score 0 to 1 at 90 days was a responder (good outcome), or (2) if the premorbid mRS score was 2, then the mRS score 0 to 2 was a responder (good outcome). The mRS score was assessed at 90 days, up to 120 days after randomization. Other outcomes include ordinal mRS score, mRS score 0 to 1, mRS

score 0 to 2 at 90 to 120 days, mRS return to premorbid function, and NIHSS score of 0 at day 5 or at discharge (whichever is sooner). For mRS return to premorbid function, patients were asked to self-report whether their current level of functioning at 90-day follow-up had returned to the prestroke level of functioning. These were assessed by blinded assessors via telephone interview using the Rankin focused assessment scale.⁷

- **Quality of life (QOL):** This included health-related QOL outcomes as assessed by telephone interview using the EuroQoL (European Quality of Life) and Lawton-Brody Instrumental Activities of Daily Living Scale index. EuroQoL consists of the European Quality of Life score index (EQ-5D-5L [EuroQoL 5-Dimension, 5-Level]) and a visual analog scale (VAS).^{8,9} The EQ-5D-5L is a 5-item generic measurement of health-related QOL that evaluates the respondent's view on mobility, self-care, usual activities, pain or discomfort, and anxiety or depression. Each item is a Likert scale response option that scores between 1 (presenting no problem) and 5 (extreme problem). The scores are then combined to form a 5-digit health state, which is then converted into a health utility score (EQ-5D-index) using the time trade-off algorithm for Canada.¹⁰ A health utility score (EQ-5D-index) ranges between −1 and 1, with 0 being the worst health state. In the final part of the EuroQoL (ie, the VAS), respondents are asked to mark their current health state on a scale of 0 (worst imaginable health state) to 100 (best imaginable health state). QOL outcomes were assessed by a blinded assessor at the same time as the mRS score assessment.
- **Safety:** Safety outcomes include progression of stroke, rescue endovascular stroke treatment, mortality, any hemorrhage, symptomatic intracranial hemorrhage, and peripheral bleeding requiring blood transfusion. Symptomatic intracranial hemorrhage was defined as a new intracranial hemorrhage (intracerebral, subarachnoid, intraventricular, or subdural hemorrhage) associated with clinical evidence of significant (NIHSS score of ≥ 2) neurological worsening, in which the hemorrhage was judged to be the most important cause of the neurological worsening. Progression of stroke was defined by the NIHSS score worsening of >2 points from baseline.

Statistical Analyses

Baseline characteristics were summarized using frequencies and proportions for categorical variables and means (or medians) with SDs (or interquartile range) as appropriate for continuous variables. All outcomes were compared between successful recanalization and unsuccessful recanalization groups in unadjusted (using 2×2 tables) and adjusted analyses. For adjusted analyses of binary outcomes, generalized linear models (with no offset) were used with a Poisson distribution log link and robust SE estimation to directly generate risk ratios. All effect sizes were adjusted for age, sex, baseline stroke severity, and onset-to-randomization time. We used a proportional odds model, adjusting for the same variables, to assess the mRS score across its distribution. Because the proportional odds assumption was not met, we did not report a common odds ratio (OR). We used the Wilcoxon rank-sum test and Somers D to compare the distribution of ordinal mRS score

across treatment groups and calculate a measure of ordinal association. Further, differences in outcomes in patients receiving tenecteplase versus standard of care in successful recanalization (yes) and successful recanalization (no) were reported separately using unadjusted and adjusted analyses for each of the outcomes mentioned above in an exploratory analysis.

To address potential bias inherent in defining groups by treatment-related recanalization status, a multivariable modeling with prespecified clinically relevant variables was performed to identify variables associated with successful recanalization within the intention-to-treat population. We selected predefined key clinical and imaging variables, including treatment type, that were associated with successful recanalization. Similarly, a separate multivariable regression model that included successful recanalization among other relevant clinical and imaging variables was used to predict early neurological outcome, defined as an NIHSS score of 0 at day 5 or hospital discharge. All reported *P* values were 2-sided, with $P < 0.05$ considered statistically significant. Statistical analysis was performed using Stata/MP version 18.0 (StataCorp LP)¹¹

RESULTS

Of the 886 patients randomized in the TEMPO-2 randomized controlled trial, 517 (58.3%) with baseline CTA evidence of occlusion and a follow-up CTA, were included in the analysis. Of these included patients, 178 (34.6%, 122: tenecteplase, 56: control) were in the successful recanalization group and 336 (65.4%, 134: tenecteplase, 202: control) were in the unsuccessful recanalization group (Figure S1, 368 excluded for no follow-up CTA needed as per RCT protocol and 1 excluded for no follow-up imaging). There were no differences in baseline characteristics of patients who had repeat CTA versus those who did not have repeat CTA, except for higher rates of vessel occlusion (large vessel occlusion 16.7% versus 4.7%, medium vessel occlusion 75.7% versus 24.9%, respectively; Table S1).

Baseline patient characteristics by the prespecified successful recanalization status are shown in Table 1. Patients with successful recanalization, as compared with unsuccessful recanalization, had similar baseline characteristics except that those patients in the successful recanalization group had lower vertebrobasilar occlusions as compared with the unsuccessful recanalization group (3.4% versus 9.8%), and successful recanalization was significantly higher in the tenecteplase arm as compared with the control arm (68.5% versus 39.9%; Table 1). Overall, successful recanalization was achieved in 122 (23.6%) in the tenecteplase arm and 56 (10.8%) in the control arm. Rates of successful recanalization by treatment type in the control arm are provided in Table S2.

Association Between Recanalization Status and Outcomes

Primary functional outcome of return to baseline neurological functioning was significantly higher in the

Table 1. Baseline Characteristics in Patients by Successful Recanalization on Follow-Up CTA (Yes/No)

Baseline characteristics	Successful recanalization	Unsuccessful recanalization	P value
N	178 (34.6%)	336 (65.4%)	
Demographics			
Age, y, median (IQR)	70 (60–80)	71 (61–80)	0.45
Sex (female), n (%)	81 (45.5%)	132 (39.3%)	0.19
Race, n (%)			
White	153 (86.0%)	278 (82.7%)	0.19
Asian	18 (10.1%)	35 (10.4%)	
Black	5 (2.8%)	5 (1.5%)	
First nations	0 (0.0%)	7 (2.1%)	
Pacific islander	0 (0.0%)	1 (0.3%)	
Other	2 (1.1%)	10 (3.0%)	
Ethnicity, n (%)			
Hispanic	175 (98.3%)	329 (97.9%)	1
Non-Hispanic	3 (1.7%)	7 (2.1%)	
Medical history, n (%)			
Hypertension	105 (59.0%)	189 (56.2%)	0.57
Past smoking	77 (43.3%)	131 (39.0%)	0.39
Hyperlipidemia	69 (38.8%)	118 (35.1%)	0.44
Diabetes	26 (14.6%)	57 (17.0%)	0.53
Past stroke	22 (12.4%)	63 (18.8%)	0.08
Atrial fibrillation	42 (23.6%)	70 (20.8%)	0.50
Ischemic heart disease	26 (14.6%)	50 (14.9%)	1
Congestive heart failure	5 (2.8%)	13 (3.9%)	0.62
Chronic renal failure	8 (4.5%)	11 (3.3%)	0.47
Peripheral vascular disease	5 (2.8%)	14 (4.2%)	0.6
Past intracerebral hemorrhage	0 (0.0%)	3 (0.9%)	0.55
Clinical presentation, median (IQR)			
NIHSS score at baseline	2 (1–3)	2 (1–3)	0.33
Hemoglobin, g/L	141 (130–151)	142 (131–151)	0.81
Glucose, mM	6 (5–7)	6 (6–8)	0.56
Creatinine, mM	86 (71–103)	82 (69–95)	0.005
Onset-to-randomization time, min	255 (152–431)	288 (166–4387)	0.82
Onset-to-hospital-arrival time, min	127 (58–301)	148 (74–326)	0.52
Onset-to-treatment time, min	266 (162–450)	307 (181–460)	0.34
Imaging features			
ASPECTS baseline	10 (10–10)	10 (9–10)	0.17
Occlusion location at baseline			
LVO	27 (15.2%)	59 (17.6%)	0.01
MeVO	145 (81.5%)	245 (72.6%)	
VB	6 (3.4%)	33 (9.8%)	
Treatment allocation, n (%)			
Control	56 (31.5%)	202 (60.1%)	<0.001
Tenecteplase	122 (68.5%)	134 (39.9%)	

ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; CTA, computed tomography angiography; LVO, large vessel occlusion; MeVO, medium vessel occlusion; NIHSS, National Institutes of Health Stroke Scale; and VB, vertebrobasilar occlusion.

successful recanalization group (143 [80.3%]) as compared with the unsuccessful recanalization group (223 [66.4%]; adjusted risk ratio, 1.21 [95% CI, 1.09–1.34]; Table 2). These results were consistent across all the other functional outcomes (mRS score, 0–1, mRS score, 0–2, mRS return to premorbid function at 90 days, and NIHSS score of 0 at day 5 or on discharge, whichever is sooner; Table 2; Figure). The overall EQ-5D-5L index was significantly higher in successful recanalization as compared to unsuccessful recanalization groups (mean, 0.86 [19.7] versus 0.79 [22.15], respectively, adjusted-difference, 0.06 [95% CI, 0.02–0.11]). EQ-5D-5L VAS ratings were also significantly higher in the successful recanalization versus the unsuccessful recanalization group (mean 77.58 versus 71.87; adjusted-difference, 5.48 [95% CI, 1.48–9.48]). There was no difference in the Lawton-Brody Instrumental Activities of Daily Living Scale index and mRS score at 90 days between 2 groups (Table 2). For safety outcomes, stroke progression was significantly lower in the successful recanalization group as compared with the unsuccessful recanalization group (2.8% versus 13.1%, adjusted risk ratio, 0.21 [95% CI, 0.08–0.52]). Other safety outcomes, such as rescue endovascular treatment, recurrent stroke, and mortality, were numerically lower but not statistically significant in the successful recanalization group as compared with the unsuccessful recanalization group (Table 3).

Predictors of Successful Recanalization

In the multivariable logistic regression model predicting successful recanalization, treatment with tenecteplase

was the strongest independent predictor (OR, 3.48 [95% CI, 2.33–5.18]). Other factors associated with lower odds of successful recanalization included resolved symptoms at treatment time (OR, 0.50 [95% CI, 0.26–0.94]) and history of prior stroke (OR, 0.55 [95% CI, 0.31–0.97]). Age, baseline NIHSS score, baseline Alberta Stroke Program Early CT Score, sex, time from onset to treatment, presence of large vessel occlusion, hyperlipidemia, diabetes, and hypertension were not significantly associated with recanalization success in the adjusted model (Table 4). In a sensitivity analysis excluding treatment assignment from the multivariable model, the magnitude and direction of the associations for the remaining predictors were largely unchanged, suggesting that inclusion of tenecteplase in the primary model did not materially influence the estimated effects of other baseline variables (Table S3).

Predictors of NIHSS Score 0 at Day 5 or Discharge

In the multivariable model predicting early neurological recovery, defined as NIHSS score of 0 at day 5, successful recanalization was the strongest predictor (OR, 3.26 [95% CI, 2.12–5.01]). Baseline stroke severity was also significantly associated with early recovery, with a higher baseline NIHSS score reducing the odds of an NIHSS score of 0 at day 5 (OR, 0.50 [95% CI, 0.36–0.70]). Other clinical variables, including age, sex, baseline Alberta Stroke Program Early CT Score, history of hypertension, hyperlipidemia, diabetes, prior stroke, presence of large vessel occlusion, time from symptom onset to treatment, and treatment assignment (tenecteplase versus control),

Table 2. Functional and Quality of Life Outcomes in Patients by Successful Recanalization 4 to 8 Hours After Randomization

	Successful recanalization	Unsuccessful recanalization	Risk difference (95% CI) OR risk ratio (unadjusted; 95% CI)	Risk ratio (adjusted*; 95% CI)
Primary functional outcome				
Responder	143 (80.3%)	223 (66.4%)	1.21 (1.09 to 1.34)	1.21 (1.09 to 1.34)
Secondary functional outcome				
mRS score 0–1 at 90 d	137 (77.0%)	215 (64.2%)	1.20 (1.07 to 1.34)	1.20 (1.07 to 1.34)
mRS score 0–2 at 90 d	159 (89.3%)	268 (80.0%)	1.12 (1.04 to 1.20)	1.11 (1.03 to 1.19)
NIHSS score 0 at 5 d or discharge	123 (69.1%)	144 (43.0%)	1.61 (1.37 to 1.88)	1.58 (1.36 to 1.84)
mRS return to premorbid function	104 (58.4%)	143 (42.6%)	1.37 (1.15 to 1.64)	1.37 (1.15 to 1.64)
Median mRS score at 90 d	1 (0 to 1)	1 (0 to 2)	...	...
Quality of life outcomes				
Mean Lawton IADL percent functioning (n=850)	90.79	87.21	3.59 (–1.06 to 8.23)†	3.66 (–0.83 to 8.15)‡
Mean EQ-5D-5L index (n=845)	0.86 (19.74)	0.79 (22.15)	0.06 (0.02 to 0.11)†	0.06 (0.02 to 0.11)‡
Mean EQ-5D-5L VAS (n=839)	77.58	71.87	5.71 (1.73 to 9.69)†	5.48 (1.48 to 9.48)‡

The data that violate the proportional odds assumptions and so an ordinal shift analysis are not presented. EQ-5D-5L indicates EuroQol 5-Dimension, 5-Level; EQ-5D-5L VAS, EQ-5D-5L visual analog scale health score; Lawton IADL, Lawton-Brody Instrumental Activities of Daily Living Scale index; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; and OR, odds ratio.

*Adjusted for age, sex, NIHSS score at baseline, and onset-to-randomization time.

†Unadjusted risk difference of means.

‡Adjusted risk difference of means.

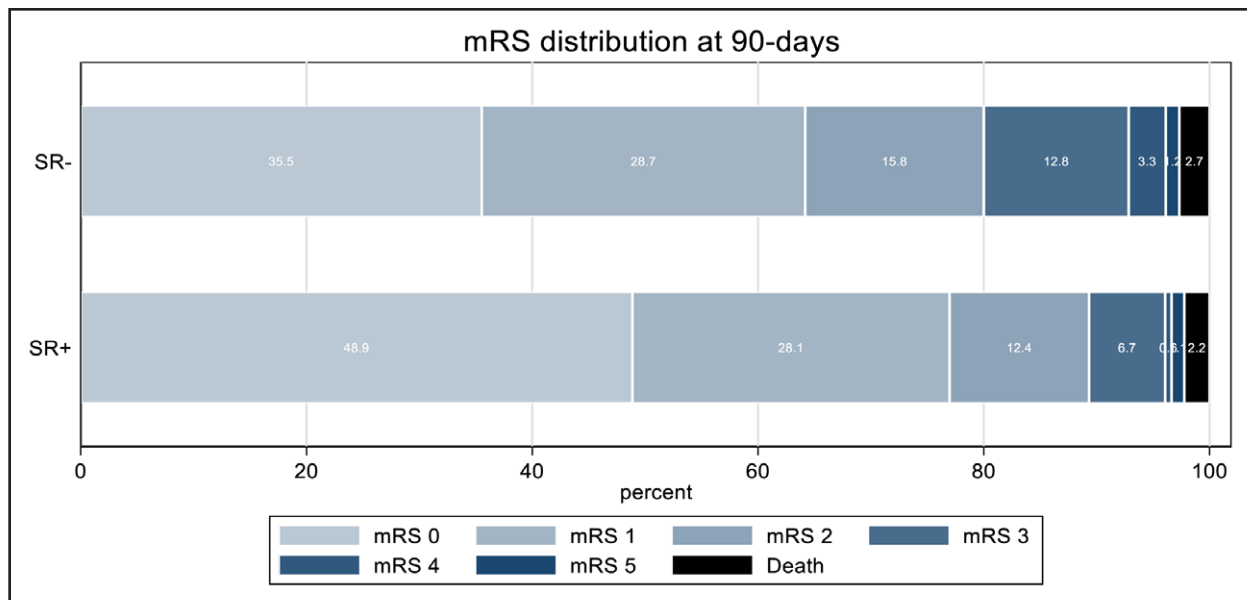


Figure. Distribution of the modified Rankin Scale (mRS) scores at 90 days by successful recanalization on follow-up computed tomography angiography (yes/no).

Scores range from 0 to 6, with 0 indicating no symptoms, 1 no clinically significant disability, 2 slight disability, 3 moderate disability, 4 moderately severe disability, 5 severe disability, and 6 death. SR+ indicates successful recanalization (yes); and SR-, successful recanalization (no).

were not significantly associated with early neurological recovery (Table 5).

Comparison by Treatment Type (Tenecteplase Versus Control) in the Successful Recanalization+Group

In the successful recanalization group, baseline characteristics were similar in the 2 treatment arms except that patients in tenecteplase arm were significantly older (median age, 70 versus 68 years, $P=0.007$) and had

more comorbidities such as hypertension (67.2% versus 41.1%, $P=0.002$), type II diabetes (67.2% versus 41.1%, $P=0.001$) and dyslipidemia (46.7% versus 21.4%, $P=0.02$), as compared with control arm (Table S4).

Rates of primary outcome of return to baseline neurological functioning were seen in 99/122 (77.9%) in the tenecteplase arm and 48 of 56 (85.7%) in the control arm (adjusted risk ratio, 0.94 [95% CI, 0.82–1.08]; Table S5). Other functional outcomes, such as mRS score 0 to 1, mRS score 0 to 2 at 90 days, median mRS score at 90 days, and NIHSS score of 0 at day 5 or discharge,

Table 3. Safety Outcomes in Patients by Successful Recanalization on Follow-Up CTA at 4 to 8 Hours After Randomization

Safety outcomes	Successful recanalization	Unsuccessful recanalization	Risk ratio or hazard ratio (unadjusted; 95% CI)	Risk ratio, or hazard ratio (adjusted*; 95% CI)
	178 (34.6%)	336 (65.4%)		
SAE occurred	27 (15.2%)	94 (28.0%)	0.54 (0.37–0.80)	0.54 (0.37–0.80)
Stroke progression	5 (2.8%)	44 (13.1%)	0.21 (0.09–0.53)	0.21 (0.08–0.52)
Stroke recurrence	5 (2.8%)	14 (4.2%)	0.67 (0.25–1.84)	0.69 (0.26–1.86)
Symptomatic ICH	1 (0.6%)	2 (0.6%)	0.94 (0.09–10.36)	1.12 (0.10–12.57)
Rescue EVT for index stroke	3 (1.7%)	13 (3.9%)	0.44 (0.13–1.51)	0.41 (0.12–1.48)
Death within 5 d	2 (1.1%)	2 (0.6%)	1.89 (0.27–13.41)	2.02 (0.27–15.19)
Death at 90 d	4 (2.2%)	9 (2.7%)	0.83 (0.26–2.69)†	0.87 (0.27–2.84)†

sICH was defined as a new intracranial hemorrhage (intracerebral, subarachnoid, intraventricular, or subdural hemorrhage) associated with clinical evidence of significant (NIHSS score ≥ 2) neurological worsening, in which the hemorrhage was judged to be the most important cause of the neurological worsening. Progression of stroke was defined by the NIHSS score worsening of >2 points from baseline. CTA indicates computed tomography angiography; EVT, endovascular thrombectomy; ICH, intracranial hemorrhage; NIHSS, National Institutes of Health Stroke Scale; SAE, serious adverse event; and sICH, symptomatic intracranial hemorrhage.

*Adjusted for age, sex, NIHSS score at baseline, and onset-to-randomization time.

†Hazard ratios.

Table 4. Multivariable Predictors of Successful Recanalization on Follow-Up CTA

Variable name	Odds ratio	Lower bound 95% CI	Upper bound 95% CI
Age, y	0.79	0.59	1.06
Men (vs women)	1.35	0.91	2.02
Type II diabetes	0.65	0.37	1.16
Hypertension	0.80	0.51	1.24
Baseline NIHSS score	0.71	0.51	1.00
Prior stroke (yes)	0.55	0.31	0.97
Dyslipidemia	1.14	0.73	1.79
Baseline ASPECTS	1.25	0.91	1.72
Onset-to-treatment time	0.99	0.75	1.31
Tenecteplase	3.48	2.33	5.18
Symptoms resolved (yes)	0.50	0.26	0.94
Large vessel occlusion (yes)	0.78	0.45	1.35

This table presents the results of a multivariable logistic regression model identifying factors associated with successful recanalization (defined as rAOL $\geq 2b/3$) on 4–8-hour follow-up CTA in patients with minor ischemic stroke and baseline intracranial occlusion enrolled in the TEMPO-2 trial (Tenecteplase Versus Standard of Care for Minor Ischemic Stroke With Proven Occlusion). Variables included in the model were selected using clinical relevance. Time from onset to treatment was imputed in 6 missing cases using multiple imputation. Adjusted OR and 95% CIs are reported. ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; CTA, computed tomography angiography; NIHSS, National Institutes of Health Stroke Scale; OR, odds ratio; and rAOL, revised Arterial Occlusive Lesion score.

were similar in both treatment arms (Table S5). QOL outcomes (EQ-5D-5L, EQ-VAS [EuroQol visual analog scale], and Lawton-Brody Instrumental Activities of Daily Living Scale index) and all safety outcomes were similar in both treatment arms (Tables S5 and S6).

Comparison by Treatment Type (Tenecteplase Versus Control) in the Unsuccessful Recanalization Group

In the unsuccessful recanalization group, baseline characteristics were similar in the 2 treatment arms (Table S4). Rates of primary outcome of return to baseline neurological functioning were seen in 84 of 134 (62.7%) in the tenecteplase arm and 139 of 202 (68.8%) in the control arm (adjusted risk ratio, 0.91 [95% CI, 0.78–1.07]; Table S5). Other functional outcomes, such as mRS score 0 to 1, mRS score 0 to 2 at 90 days, median mRS score at 90days, NIHSS score of 0 at day 5 or discharge, were similar in both treatment arms (Table S5). QOL outcomes (EQ-5D-5L, EQ-VAS, and Lawton-Brody Instrumental Activities of Daily Living Scale index) and all safety outcomes were similar in both treatment arms (Tables S5 and S6).

DISCUSSION

This subgroup analysis of the neutral TEMPO-2 study yielded 2 major findings. First, successful recanalization

Table 5. Multivariable Predictors Early Neurological Recovery (NIHSS Score of 0 at Day 5/Discharge)

Variable name	Odds ratio	Lower bound 95% CI	Upper bound 95% CI
Age, y	1.09	0.82	1.46
Men (vs women)	0.93	0.63	1.38
Hypertension	1.27	0.82	1.97
Dyslipidemia	1.20	0.77	1.87
Type II diabetes	0.89	0.52	1.52
Prior stroke	1.11	0.66	1.87
Baseline NIHSS score	0.50	0.36	0.70
Onset-to-treatment time	1.02	0.78	1.34
Baseline ASPECTS	1.20	0.89	1.63
Symptoms resolved (yes)	1.87	0.98	3.56
Tenecteplase	0.95	0.64	1.42
Large vessel occlusion	1.26	0.74	2.13
Successful recanalization	3.26	2.12	5.01

This table displays the adjusted odds ratios for achieving a NIHSS score of 0 at day 5 or discharge variables included in the model were selected using clinical relevance, and adjusted OR with 95% CIs are shown. ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; NIHSS, National Institutes of Health Stroke Scale; and OR, odds ratios.

led to improved early and day 90 functional and QOL outcomes, regardless of treatment modality. The findings support the notion that more complete restoration of cerebral perfusion has an impact on both recovery and QOL in patients with stroke, even in patients presenting with minor nondisabling stroke. Second, contemporary medical therapy may facilitate meaningful levels of delayed or endogenous recanalization (10% in the current study at 4–8 hours of repeat CTA). This highlights an important evolution in stroke care: with advances in medical management, a substantial proportion of patients can still achieve favorable outcomes, particularly if recanalization, whether pharmacologically induced or spontaneous, occurs within a relevant therapeutic window. Importantly, this study provides the first clinical trial data on recanalization rates achieved with contemporary standard care, offering a valuable benchmark for future research and informing treatment decisions in minor stroke with occlusion.

These findings align with prior studies showing the efficacy of tenecteplase in promoting recanalization in ischemic stroke, including the EXTEND-IA TNK (Tenecteplase Versus Alteplase Before Thrombectomy for Ischemic Stroke) and TIMELESS (Tenecteplase for Stroke at 4.5 to 24 Hours With Perfusion-Imaging Selection) trials.^{12–14} The ability of tenecteplase to improve recanalization rates in patients with minor ischemic stroke and intracranial occlusion further underscores its potential as an effective early intervention in this specific population. In the current study, multivariable analysis identified tenecteplase treatment as the strongest independent predictor of successful recanalization, aligning with prior studies that have demonstrated the superior thrombolytic efficacy of tenecteplase compared with

alteplase or standard care in acute ischemic stroke populations.^{14–16} The association of lower baseline NIHSS score and resolved symptoms with higher recanalization likelihood is consistent with evidence suggesting that less severe strokes may have smaller clot burden or more favorable collateral circulation, facilitating reperfusion.¹⁷

Although our analysis showed that those who received repeat CTA were broadly similar in age, baseline NIHSS score, and onset-to-treatment time compared with those who did not, aside from the expected difference in presence of baseline occlusion— $\approx 30\%$ of patients with confirmed occlusion did not undergo follow-up vascular imaging. The reasons for this were often logistical (eg, overnight imaging limitations in certain regions), potentially introducing site- or region-specific biases into who contributed to the imaging subgroup. The measurement of recanalization was based on a single time point (4–8 hours), and it is possible that late recanalization, facilitated by dual antiplatelet therapy and endogenous thrombolysis, occurred in both arms, ultimately diminishing the impact of early recanalization on functional outcomes. Prior studies have shown that spontaneous recanalization can occur even in the absence of thrombolytic therapy, particularly in minor stroke with slow infarct evolution.¹⁸ We observed that the rate of stroke progression was significantly lower in the successful recanalization group, likely due to early reperfusion preventing infarct expansion and preserving penumbral tissue, but without an increase in risk of any safety outcomes or reperfusion injury. These results also provide reassurance about the safety profile of tenecteplase in patients with minor ischemic stroke and suggest that the benefit of early recanalization is not accompanied by an increased risk of thrombolytic-related serious adverse events. Finally, the assessment of recanalization may occur in the presence of reduced cerebral flow in the microcirculation, affecting tissue viability and, thereby, functional recovery. Safety data from this analysis were consistent with previous studies evaluating intravenous thrombolytics.^{19–21} Potential explanations include that patients who fail to recanalize may represent a biologically distinct, higher-risk subgroup (eg, more resistant clot), whereas delayed or spontaneous recanalization, potentially facilitated by endogenous fibrinolysis and contemporary medical therapy, including routine dual antiplatelet treatment, may attenuate differences between treatment arms. Clot location did not emerge as a predictor of successful recanalization in our analysis. In addition, early recanalization, while associated with reduced stroke progression in our study, may not always translate into measurable differences in 90-day functional outcomes in a population with relatively mild baseline deficits and high rates of recovery. These findings underscore the complexity of the relationship between treatment, recanalization, and clinical outcomes in minor stroke and highlight the need for further mechanistic studies.

In this subgroup analysis, tenecteplase was the strongest independent predictor of early successful recanalization, and successful recanalization was the strongest predictor of improved 90-day functional outcome. Despite this biological relationship, tenecteplase did not confer a statistically significant overall functional benefit compared with standard care, mirroring the neutral results of the TEMPO-2 trial.⁵ Several factors may explain this apparent paradox. First, the rate of symptomatic intracranial hemorrhage was low and comparable between groups, suggesting that safety events did not offset the potential benefit of early recanalization. Second, the widespread and rapid initiation of dual antiplatelet therapy in contemporary stroke practice, now routine following CHANCE (Clopidogrel in High-Risk Patients With Acute Non-Disabling Cerebrovascular Events) and POINT (Platelet-Oriented Inhibition in New TIA and Minor Ischemic Stroke) trials, may facilitate later spontaneous or pharmacologically mediated recanalization in control patients, narrowing between-group differences by 24 hours.^{22,23} Third, in patients with minor, nondisabling strokes and preserved tissue, later recanalization may still yield favorable outcomes, reducing the incremental benefit of earlier reperfusion. Fourth, outcome measurement in this low NIHSS score population may be limited by ceiling effects of the mRS score, making small functional gains difficult to detect.²⁴ Importantly, we found no heterogeneity in treatment effect by age, sex, baseline NIHSS score, onset-to-treatment time, occlusion site, or large vessel occlusion status, suggesting the subgroup was broadly representative of the trial population. Approximately 30% of patients with a baseline occlusion did not undergo repeat CTA, largely due to logistical factors such as scanner access, interhospital transfers, or off-hours imaging limitations. These operational considerations and unmeasured confounders may have influenced the observed associations. Together, these findings highlight that while tenecteplase substantially increases early recanalization, modern best medical therapy, including dual antiplatelet therapy, may allow later recanalization in control patients, attenuating treatment-level differences in long-term outcomes.

We acknowledge the limitations of this study. First, the findings are based on subgroup analyses within the neutral TEMPO-2 randomized controlled trial, and therefore, the sample size may not be adequately powered to detect small differences. Second, given that multiple comparisons were made, there is a risk of type I error (false positives). Some associations identified in the study were possibly due to chance rather than true effects. The subgroup analysis is only hypothesis-generating, and large confidence limits may be due to sparse-data bias.²⁵ Third, this is a post hoc analysis, and only those subjects with occlusions and follow-up imaging that could have led to some selection bias. Lastly, the study did not explore other potential factors that could influence outcomes,

such as ischemia severity, clot burden, and final infarct volume on follow-up scan.

In conclusion, our analysis of the TEMPO-2 trial supports the hypothesis that successful early recanalization is associated with improved functional outcomes and reduced neurological deterioration in patients with minor ischemic stroke with intracranial occlusion. Intravenous tenecteplase was shown to increase the likelihood of successful recanalization compared with standard care, but only a minority recanalized, and no significant differences were observed in functional outcomes between the 2 treatment groups once recanalization was achieved. These findings underscore the need to further enhance early recanalization rates in this patient population and suggest that successful recanalization, rather than the treatment type, may be the key factor in determining long-term outcomes.

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Affiliations

Department of Internal Medicine (Neurology Division), University of Manitoba, Health Sciences Center, Winnipeg, Canada (N.S.). Department of Neurology, Helsinki University Hospital and University of Helsinki, Finland (D.S.). Department of Community Health Sciences (S.V., P.A.B., M.D.H., S.B.C.), Department of Clinical Neurosciences (S.V., A.G., P.A.B., M.D.H., S.B.C.), and Department of Radiology (P.A.B., S.B.C.), University of Calgary, Canada. Department of Neurosciences, University of British Columbia, Vancouver, Canada (T.S.F.). Department of Medicine (Neurology), University of Toronto, Sunnybrook Health Sciences Centre, Canada (A.Y.X.Y.). Department of Neuroscience, Eastern Health, Eastern Health Clinical School, Monash University, Victoria, Australia (P.M.C.C.). Division of Neurology, Department of Medicine, University of Alberta, Edmonton, Canada (B.H.B.). Department of Neurology, Royal Adelaide Hospital, Adelaide, South Australia, Australia (T.J.K.). Department of Medicine and Neurology, Melbourne Brain Centre at the Royal Melbourne Hospital, University of Melbourne, Parkville, Victoria, Australia (B.C.). Department of Neurology, Vall D'hebron University Hospital, Vall D'hebron Research Institute, Barcelona, Spain (C.M.). Department of Neurology, University of Glasgow, Scotland, United Kingdom (K.W.M.).

Author Contributions

Dr Singh prepared the first draft of the report. Drs Singh, Strbian, Coutts, and Hill conceptualized the study design. S. Vatanpour did the initial analysis with Drs Singh, Coutts, and Hill providing additional supervision, data management, and statistical support. All authors participated in patient enrollment, trial execution, management, critical review of the report, and approval of the final version before submission.

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Supplemental Material

Tables S1–S6

Figure S1

STROBE Checklist

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